

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**_____
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[\sqrt{2} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[\sqrt{2} \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$

Find the quotient. Write it in rectangular form, using exact values. (3 points)

2. $\frac{\sqrt{2} + i\sqrt{6}}{-\sqrt{3} - i}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $\left[\sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^4$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4. $x^4 + 8 + 8i\sqrt{3} = 0$

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute z , $z^2 + z$, $(z^2 + z)^2 + z$, $[(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a) $z = -2$

YES / NO

b) $z = -1 + i$

YES / NO

ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Find the product. Write the product in rectangular form, using exact values. (3 points)

$$1. \left[\sqrt{2} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[\sqrt{2} \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$$

WORK

$$\begin{aligned} &= \sqrt{2} \cdot \sqrt{2} \operatorname{cis} \left(\frac{5\pi}{4} + \frac{11\pi}{12} \right) \\ &= 2 \operatorname{cis} \left(\frac{15\pi}{12} + \frac{11\pi}{12} \right) \\ &= 2 \operatorname{cis} \frac{26\pi}{12} = 2 \operatorname{cis} \frac{13\pi}{6} \\ &= 2 \operatorname{cis} \frac{\pi}{6} \\ &= 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \\ &= 2 \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) \\ &= \boxed{\sqrt{3} + i} \end{aligned}$$

Trap: given in radians, stay in radians.

WHY

Product theorem: multiply the r 's, add the θ 's.

Common denominator: $\frac{5\pi}{4} = \frac{15\pi}{12}$.

Simplify.

$\frac{13\pi}{6} - \frac{12\pi}{6} = \frac{\pi}{6}$ (subtract 2π).

Write cis out.

Unit circle at $\frac{\pi}{6}$.

Find the quotient. Write it in rectangular form, using exact values. (3 points)

$$2. \frac{\sqrt{2} + i\sqrt{6}}{-\sqrt{3} - i}$$

WORK

$$r_1 = \sqrt{(\sqrt{2})^2 + (\sqrt{6})^2} = \sqrt{2+6} = 2\sqrt{2}$$

$$\theta_1 = \tan^{-1} \left(\frac{\sqrt{6}}{\sqrt{2}} \right) = \tan^{-1}(\sqrt{3}) = 60^\circ$$

$$r_2 = \sqrt{(-\sqrt{3})^2 + (-1)^2} = \sqrt{3+1} = 2$$

$$\theta_2 = \tan^{-1} \left(\frac{-1}{-\sqrt{3}} \right) = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = 30^\circ$$

$$\theta_2 = 30^\circ + 180^\circ = 210^\circ$$

$$\frac{2\sqrt{2} \operatorname{cis} 60^\circ}{2 \operatorname{cis} 210^\circ} = \frac{2\sqrt{2}}{2} \operatorname{cis} (60^\circ - 210^\circ)$$

$$= \sqrt{2} \operatorname{cis} (-150^\circ)$$

$$= \sqrt{2} \operatorname{cis} 210^\circ$$

$$= \sqrt{2} \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right)$$

$$= \boxed{-\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}i}$$

WHY

Top first. $\sqrt{8} = 2\sqrt{2}$.

Both positive: QI, no fix.

Bottom.

Trap: both negative is QIII, not 30° .

Quotient theorem: divide the r 's, subtract the θ 's.

$-150^\circ + 360^\circ = 210^\circ$.

QIII: both negative.

$\sqrt{2} \cdot \sqrt{3} = \sqrt{6}$.

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $\left[\sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^4$

WORK

$$\begin{aligned} &= (\sqrt{2})^4 \operatorname{cis} \left(4 \cdot \frac{2\pi}{3} \right) \\ &= 4 \operatorname{cis} \frac{8\pi}{3} \\ &= 4 \operatorname{cis} \frac{2\pi}{3} \\ &= 4 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) \\ &= \boxed{-2 + 2\sqrt{3}i} \end{aligned}$$

WHY

De Moivre: r^n and $n\theta$.

$$(\sqrt{2})^4 = (\sqrt{2})^2(\sqrt{2})^2 = 4, \text{ not } 4\sqrt{2}.$$

$$\frac{8\pi}{3} - \frac{6\pi}{3} = \frac{2\pi}{3} \text{ (subtract } 2\pi\text{).}$$

$$\frac{2\pi}{3} \text{ is QII: } \cos -, \sin +.$$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4. $x^4 + 8 + 8i\sqrt{3} = 0$

WORK

$$\begin{aligned} x^4 &= -8 - 8i\sqrt{3} \\ r &= \sqrt{(-8)^2 + (-8\sqrt{3})^2} = \sqrt{64 + 192} = 16 \\ \theta &= \tan^{-1} \left(\frac{-8\sqrt{3}}{-8} \right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \\ \theta &= \frac{\pi}{3} + \pi = \frac{4\pi}{3} \\ x &= \sqrt[4]{16} \operatorname{cis} \left(\frac{4\pi/3 + 2\pi k}{4} \right) \\ &= 2 \operatorname{cis} \left(\frac{\pi}{3} + \frac{\pi}{2}k \right) \\ k=0: \quad 2 \operatorname{cis} \frac{\pi}{3} &= 2 \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = 1 + \sqrt{3}i \\ k=1: \quad 2 \operatorname{cis} \frac{5\pi}{6} &= 2 \left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) = -\sqrt{3} + i \\ k=2: \quad 2 \operatorname{cis} \frac{4\pi}{3} &= 2 \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = -1 - \sqrt{3}i \\ k=3: \quad 2 \operatorname{cis} \frac{11\pi}{6} &= 2 \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = \sqrt{3} - i \\ x &= \boxed{1 + \sqrt{3}i, \quad -\sqrt{3} + i, \quad -1 - \sqrt{3}i, \quad \sqrt{3} - i} \end{aligned}$$

WHY

Isolate x^4 : both terms flip sign.

$$(8\sqrt{3})^2 = 64 \cdot 3 = 192. \quad \sqrt{256} = 16.$$

Trap: both negative is QIII: add π .

n th root theorem, $k = 0, 1, 2, 3$.

$$\sqrt[4]{16} = 2. \text{ Fourth roots sit } \frac{2\pi}{4} = \frac{\pi}{2} \text{ apart.}$$

$$\frac{\pi}{3} + \frac{\pi}{2} = \frac{2\pi}{6} + \frac{3\pi}{6} = \frac{5\pi}{6}.$$

$$\text{Check: } (2 \operatorname{cis} \frac{\pi}{3})^4 = 16 \operatorname{cis} \frac{4\pi}{3}.$$

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute z , $z^2 + z$, $(z^2 + z)^2 + z$, $[(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a) $z = -2$ YES / NO WORK a) $z = -2$: $-2 = 2$ $z^2 + z = (-2)^2 + (-2) = 4 - 2 = 2$ $2 = 2$ $(2)^2 + (-2) = 2$ YES b) $z = -1 + i$: $-1 + i = \sqrt{2} \approx 1.4$ $z^2 + z = (-1 + i)^2 + (-1 + i)$ $= (1 - 2i + i^2) - 1 + i = -1 - i$ $-1 - i = \sqrt{2} \approx 1.4$ $(-1 - i)^2 + (-1 + i) = (1 + 2i + i^2) - 1 + i$ $= 2i - 1 + i = -1 + 3i$ $-1 + 3i = \sqrt{1 + 9} = \sqrt{10} \approx 3.2$ NO	b) $z = -1 + i$ YES / NO WHY Equal to 2 doesn't exceed 2: keep going. Stuck at 2 forever, never over 2. Not over 2. $(-1 + i)^2 = 1 - 2i - 1 = -2i$. Still not over 2. Square the last result, add z. <i>Trap: it fails on the third check. Don't stop early and call it YES.</i>
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